

APPENDIX A
PROOF OF PROPOSITION 1

In this section, we prove the existence and uniqueness of NE in Game 1 separately.

A. NE Existence in Game 1

According to Debreu-Fan-Glicksberg Theorem [22], if each decision variable is a compact and convex set, and the objective function is quasi-concave in its decision variable, the game has at least one pure NE.

In Game 1, clients choose $\alpha_i \in [0, 1]$ and the server chooses $q \in [0, 1]$. These are compact and convex sets.

Further, U_i in Problem (3) is concave [23] in α_i for each client because of its second derivative is negative, as

$$\frac{\partial^2 U_i}{\partial \alpha_i^2} = -2eP_i q \leq 0. \quad (13)$$

For the server, ϕ is a linear function in q .

Therefore, we can prove that Game 1 has at least one NE.

B. NE Uniqueness in Game 1

We start with two lemmas and then complete the proof.

Lemma 4. *Given a detection probability $q \in [0, 1]$ of the server, the resulting attacking profile $\alpha(q)$ is unique.*

Proof. For $q > 0$, we invoke Rosen's uniqueness theorem for concave games [25]. A sufficient condition for a unique joint equilibrium is that the game is Diagonally Strictly Concave (DSC), which holds if $G + G^T$ is negative definite with G being the game's Jacobian matrix. In our game, the diagonal entries (self-concavity) are $G_{ii} = \frac{\partial^2 U_i}{\partial \alpha_i^2} = -2qeP_i$, and the off-diagonal entries (strategic externalities) are $G_{ij} = \frac{\partial^2 U_i}{\partial \alpha_i \partial \alpha_j} = qe\delta_i(1 - 2e\alpha_j)$ for $i \neq j$. By applying Gershgorin circle theorem to the symmetric matrix $G + G^T$, it is strictly negative definite if the following condition is satisfied:

$$|-4qeP_i| > \sum_{j \neq i} |qe\delta_i(1 - 2e\alpha_j) + qe\delta_j(1 - 2e\alpha_i)|. \quad (14)$$

Condition (14) reflects the inherent design of stealthy backdoor attacks [1], where the high-value gain of a successful trigger fundamentally dominates the marginal loss caused by systemic model degradation. That is, a client's own maximum backdoor-induced gain fundamentally outweighs the aggregate poisoning externalities (i.e., $2P_i \gg \sum_{j \neq i} \delta_i$). Thus, the matrix $(G + G^T)$ is strictly diagonally dominant when the practical condition (14) holds. Therefore, the game is strictly DSC, and $\alpha(q)$ is unique. Additionally, for $q = 0$, the utility function in (3) decouples into a linear function, trivially yielding a unique boundary response for each client. $\square$

Lemma 5. *Given the unique mapping $\alpha(q)$, the server's optimal detection probability q^* that forms a Nash Equilibrium is unique.*

Proof. Let $H(q) \triangleq y(\alpha(q))$, where $y(\alpha)$ is the server's threshold function defined in (10). According to the server's best

response in (10), the game admits three possible equilibrium regimes: $q^* = 1$ if $H(1) > 0$; $q^* = 0$ if $H(0) < 0$; or an interior equilibrium $q^* \in (0, 1)$ which requires $H(q^*) = 0$.

We only need to prove that if $H(q) = 0$ occurs, the root q^* is strictly unique, because q is unique in the other two possible equilibriums. From Lemma 1, for a profitable client, the unconstrained optimal parameter $\theta_i(q)$ strictly decreases as q increases. Because of the projection bound $[\cdot]_0^1$, the clipped optimal response $\alpha_i(q)$ is monotonically non-increasing in q . Furthermore, taking the partial derivative for client $k \in \mathcal{I}$ yields:

$$\frac{\partial y}{\partial \alpha_k} = 2\lambda e\alpha_k + e(S + c_k) - e\left(\sum_{i \in \mathcal{I}} c_i\right) \prod_{j \neq k} (1 - e\alpha_j) \geq 0. \quad (15)$$

Condition (15) reveals a practical parameter setting where the punishment S and poisoning deficit λ dominate the unlearning compensation costs. Thus, the strictly positive derivative in Condition (15) ensures $y(\alpha)$ is strictly monotonically increasing with respect to α . Consequently, the composite function $H(q)$ is monotonically non-increasing in q .

The function $H(q)$ can only possess multiple roots if it evaluates to zero and remains constant (flat) over an interval of q . $H(q)$ is only constant if the profile $\alpha(q)$ is invariant, which requires all clients to be fully projected to the boundaries (0 or 1). However, if all clients are fully deterred such that $\alpha = \mathbf{0}$, then $H(q) = y(\mathbf{0}) = -\sigma < 0$. Conversely, consider $\alpha = \mathbf{1}$, we have $H(q) = y(\mathbf{1}) = \lambda eI + e \sum_{i \in \mathcal{I}} (S + c_i) - \sigma + (\sum_{i \in \mathcal{I}} c_i)(1 - e)^I - \sum_{i \in \mathcal{I}} c_i$. We require the system to have a practical setting $y(\mathbf{1}) > 0$ to ensure that, at least in the worst-case scenario of a maximum, all-client attack ($\alpha_i = 1, \forall i$), the server's economic benefits from preventing catastrophic model degradation and collecting punishments strictly outweigh its operational detection and unlearning costs. Therefore, the continuous function $H(q)$ transitions from strictly negative to positive on the two sides, and its root lies in $(0, 1)$. For $H(q)$ to equal zero, there must be at least one active client k strictly in the interior $\alpha_k \in (0, 1)$, where $\alpha_k(q)$ strictly decreases in q . Thus, $H(q)$ strictly decreases exactly when it crosses zero, guaranteeing at most one unique root q^* . $\square$

Finally, we complete the proof for NE uniqueness in Proposition 1. By Lemma 4, every q maps to a strictly unique joint client best response $\alpha(q)$. By Lemma 5, the server's best response to this monotonic mapping yields exactly one stable detection probability q^* . Therefore, the combined strategy profile $(q^*, \alpha(q^*))$ constitutes the unique Nash Equilibrium of Game 1.

APPENDIX B
PROOF OF PROPOSITION 3

We first prove that (11) is a NE because all players in Game 1 perform their best response, given the other player's strategies.

For the server, when $\alpha_k^* = \frac{-e(S - \sum_{j \neq k} c_j) + \sqrt{e^2(S - \sum_{j \neq k} c_j)^2 + 4\lambda e\sigma}}{2\lambda e}$ and $\alpha_j^* = 0, \forall j \neq k$, its threshold function in (10) is

$$y(\alpha_k^*, \alpha_{-k}^*) = \lambda e \alpha_k^{*2} + e(S - \sum_{j \neq k} c_j) \alpha_k - \sigma = 0. \quad (16)$$

Thus, choosing $q^* \in (0, 1)$ in (11) is its best response.

For client k , applying q^* and $\alpha_j^* = 0$, we can compute

$$\theta_k(q^*, \alpha_{-k}^*) > 0. \quad (17)$$

Thus, $\alpha_k^* > 0$ holds.

For client j , applying q^* and α_k^* , we can compute

$$\theta_j(q^*, \alpha_{-j}^*) = \frac{S + \beta_k M}{2P_j} \frac{P_j - A_j}{P_k - A_k} + \frac{\delta_j(1 - e\alpha_k^*)\alpha_k^* - S - \beta_j M}{2P_j} < 0. \quad (18)$$

Thus, choosing $\alpha_j^* = 0$ is its best response.

APPENDIX C PROOF OF LEMMA 3

To prove (12), we first compute the gap between our mechanism and No-Attack in the server's payoff and clients' total utility.

A. Server's Payoff Gap

Applying the equilibrium in (11) to server's payoff in (5), we can get

$$\begin{aligned} \phi_{Ours} &= \lambda e q^* \alpha_k^{*2} + e q(S - \sum_{j \neq k} c_j) \alpha_k^* - \sigma q^* - \lambda \alpha_k^* \\ &= q^* y(\alpha^*) - \lambda \alpha_k^* \\ &= -\lambda \alpha_k^*, \end{aligned} \quad (19)$$

where the second to the third line is because of (16).

Also, we can find that $\phi_{No-Attack} = 0$ because $\alpha_i = 0, \forall i \in \mathcal{I}$ and $q = 0$ in No-Attack baseline.

B. Clients' Total Utility Gap

Applying the equilibrium in (11), we can derive the client's utility in our mechanism as

$$U_k = -e P_k q^* \alpha_k^{*2} + (P_k - A_k - e q^*(S + \beta_k M)) \alpha_k^* + \beta_k M, \quad (20)$$

and

$$U_j = e \delta_j q^* \alpha_k^{*2} - \delta_j \alpha_k^* + \beta_j M, \forall j \neq k. \quad (21)$$

Also, we can find $U_i = \beta_i M, \forall i \in \mathcal{I}$ in No-Attack baseline.

C. Social Welfare Gap

Summing up the server's payoff in (19), client's utility in (20) and (21), the social welfare in our mechanism is

$$\begin{aligned} SW_{Ours} &= e q^* (\sum_{j \neq k} \delta_j - P_k) \alpha_k^{*2} \\ &\quad + (P_k - \lambda - A_k - e q^* \beta_k M - \sum_{j \neq k} \delta_j - e S q^*) \alpha_k^* \\ &\quad + \sum_{i \in \mathcal{I}} \beta_i M. \end{aligned} \quad (22)$$

For No-Attack baseline, applying $q = 0$ and $\alpha_i = 0, \forall i \in \mathcal{I}$, we can find $SW_{No-Attack} = \sum_{i \in \mathcal{I}} \beta_i M$.

Thus, the social welfare difference between our mechanism and No-Attack baseline is

$$\begin{aligned} &SW_{No-Attack} - SW_{Ours} \\ &= -e q^* (\sum_{j \neq k} \delta_j - P_k) \alpha_k^{*2} \\ &\quad - (P_k - \lambda - A_k - e q^* \beta_k M - \sum_{j \neq k} \delta_j - e S q^*) \alpha_k^*, \end{aligned} \quad (23)$$

which is a quadratic function in α_k^* and approaches zero because $\lim_{S \rightarrow \infty} \alpha_k^* = 0$.